HDOC Day-43 – Menu-Driven C Program

Aim

To write a **menu-driven C program using separate functions** to:

1. Reverse each word in a sentence without changing the word order.
2. Print the initials of a name.
3. Exit.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

// Function to reverse characters from start to end
void reverse(char str[], int start, int end)
{
    char temp;

    while (start < end)
    {
        temp = str[start];
        str[start] = str[end];
        str[end] = temp;

        start++;
        end--;
    }
}

// Function to reverse each word of a sentence
void reverseWords(char str[])
{
    int i, start = 0;
    int len = strlen(str);

    for (i = 0; i <= len; i++)
    {
        if (str[i] == ' ' || str[i] == '\0')
        {
            reverse(str, start, i - 1);
            start = i + 1;
        }
    }

    printf("Output: %s\n", str);
}

// Function to print initials of a name
void printInitials(char name[])
{
```

```
    int i;

    printf("Initials: ");

    if (name[0] != '\0')
    {
        printf("%c.", toupper(name[0]));
    }

    for (i = 1; name[i] != '\0'; i++)
    {
        if (name[i - 1] == ' ' && name[i] != ' ')
        {
            printf("%c.", toupper(name[i]));
        }
    }

    printf("\n");
}

int main()
{
    int choice;
    char str[200];

    do
    {
        printf("\n===== HDOC DAY-44 =====\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(str, sizeof(str), stdin);
                str[strcspn(str, "\n")] = '\0';

                reverseWords(str);
                break;

            case 2:
                printf("Enter a name: ");
                fgets(str, sizeof(str), stdin);
                str[strcspn(str, "\n")] = '\0';

                printInitials(str);
                break;

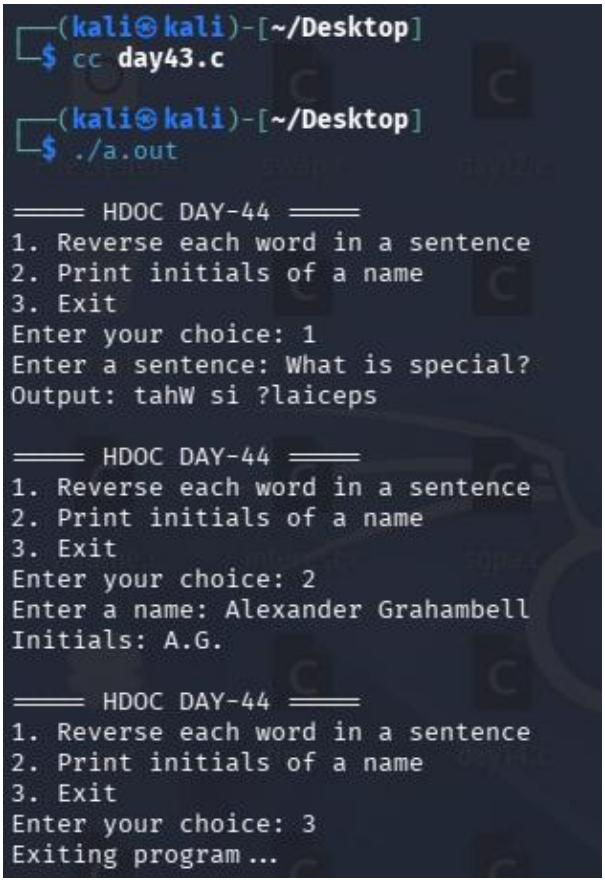
            case 3:
                printf("Exiting program...\n");
                break;
```

```
            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Sample Outputs for the 3 choices –



Functions Used

Function	Purpose
`reverse()`	Reverses characters of one word
`reverseWords()`	Finds and reverses every word separately
`printInitials()`	Prints the first letter of each part of the name

Function	Purpose
`main()`	Displays the menu and calls the required function

Result

Thus, the **menu-driven C program** to reverse each word of a sentence and print the initials of a name using **separate user-defined functions** was successfully implemented.